small electrical charge which is converted one pixel at a time to voltage, and later converted into digital information. A CMOS imaging chip has more functions than this, employing the "active pixel sensor" which allows for additional circuitry to be included on the chip itself for the conversion of voltage into digital data.

Both technologies manage to produce the highest image quality though, with no obvious advantage in one over the other, but the basic difference is that the CMOS integrated circuits have potentially more processing functions on the chip itself. On the downside, when a CCD sensor is overloaded, the bright light may cause vertical smudge, though high-end frame transfer ones have gotten rid of this problem. On the other hand, CMOS sensors have rolling shutters, typically undesirable on their own. On the plus side, CMOS chips are not only more economical to manufacture but also tend to use less power, generate less heat and provide faster read-out of the electrical signal data, while providing additional signal processing such as noise reduction on the chip itself. This explains to some extent why CMOS-chipped cameras tend to provide more highly processed, less "raw" data than the CCD chips normally found in medium format backs. In comparison, CCD is a more mature technology providing equal results as far as image quality is concerned. There are hybrid sensors, using components of both, CCD and CMOS chips in the research phase that can potentially harness benefits of both technologies in the future.

Let's look a bit into some popular colour filter technologies commonly used beginning with U.S. Patent Number 3,971,065, belonging to Bryce Bayer, received in 1976, leading to present day Bayern pattern filter mosaics. His terminology meant calling the green photo-sensors luminance-sensitive elements and the red and blue ones chrominance-sensitive elements. To mimic luminance as perceived by the retinas of human eyes, he used twice as many green elements as red or blue. In our eyes, several sensor elements, sensels, pixel sensors or simply pixels, responsible for luminance perception, use M and L cone cells, and after some interpolation, sample values as perceived by them become image pixels. Data from a single pixel can't be used on its own to determine colour since each pixel is filtered to record only one of three colours. Various demosaicing algorithms are put in place to interpolate groups of complete red, green and blue values for each point in order to obtain a full-colour image.

Almost all digital cameras use the Bayer filter only, though there exist common alternatives such as the Cyan-Yellow-Green-Magenta (CYGM)